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TECHNICAL NOTE · POWER QUALITY

Phase unbalance, voltage and power factor: what really matters

Research checked against real meter data from a 20 kWp zero-export workshop in Chiang Rai · Solar ESS · Rev 01 · 30 October 2026

72–86%
current unbalance between phases on normal days at this site

1.7–2.0%
average daytime voltage unbalance; peaks of 4.5–5.2%

1.00
vector power factor of a purely resistive load, however unbalanced

0.85
PEA power factor threshold: kVAR above 61.97% of kW is charged

The short answer

Unbalance does not create a PF charge by itself

PEA bills reactive power on the three-phase total. An unbalanced but resistive load still has a power factor of 1.00. Only reactive loads such as motors and compressors create kVAR.

It does raise voltage unbalance

Uneven current makes the heavy phase's voltage sag. At this site the daytime voltage unbalance reaches 2%, the supply limit, with peaks above 4%. Three-phase motors run hotter.

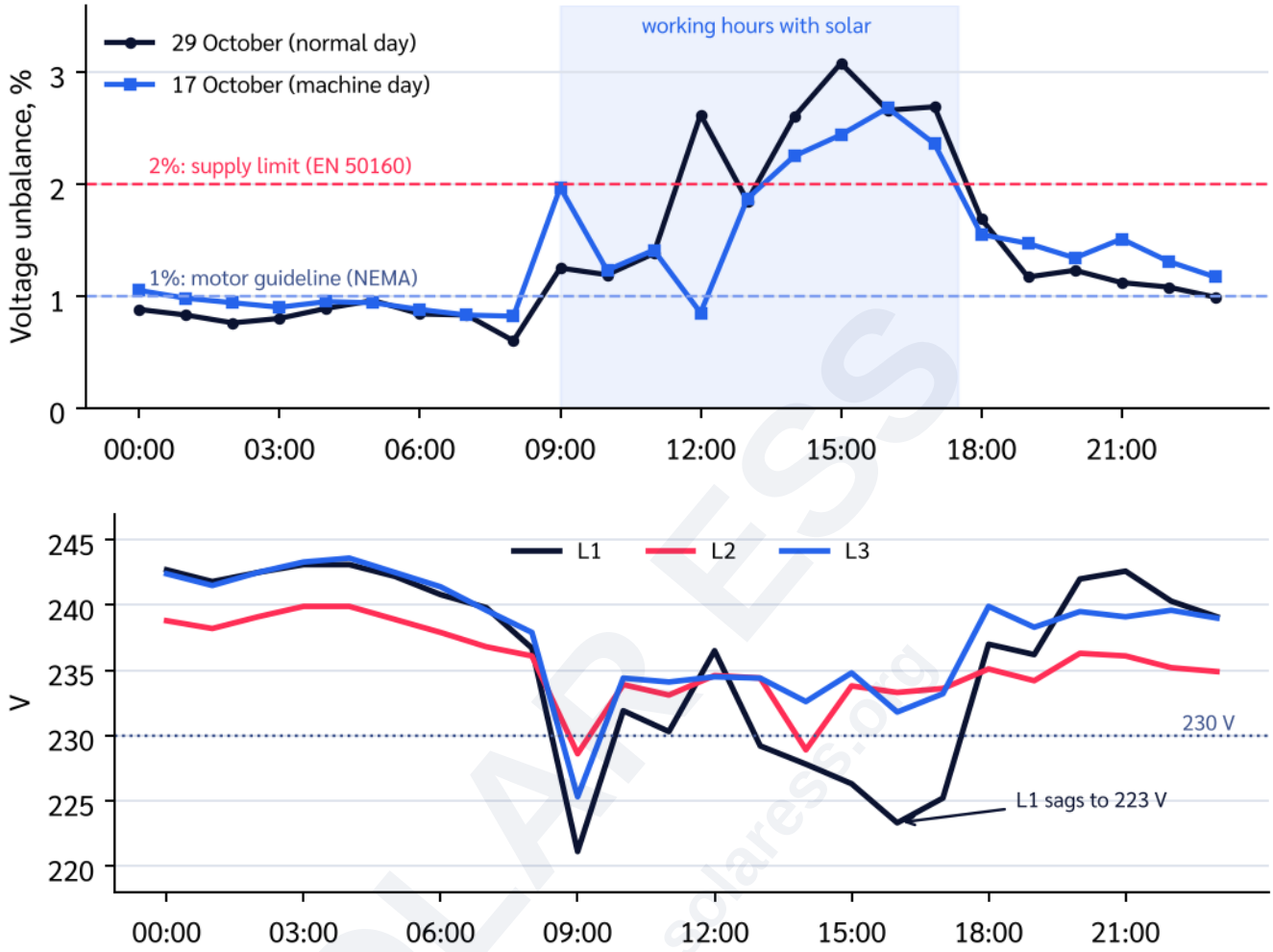
The cheapest fix is the same

Move single-phase loads from the heavy phase. Check the PEA bill and any capacitor bank before buying power-quality equipment.

1 Common claims, checked

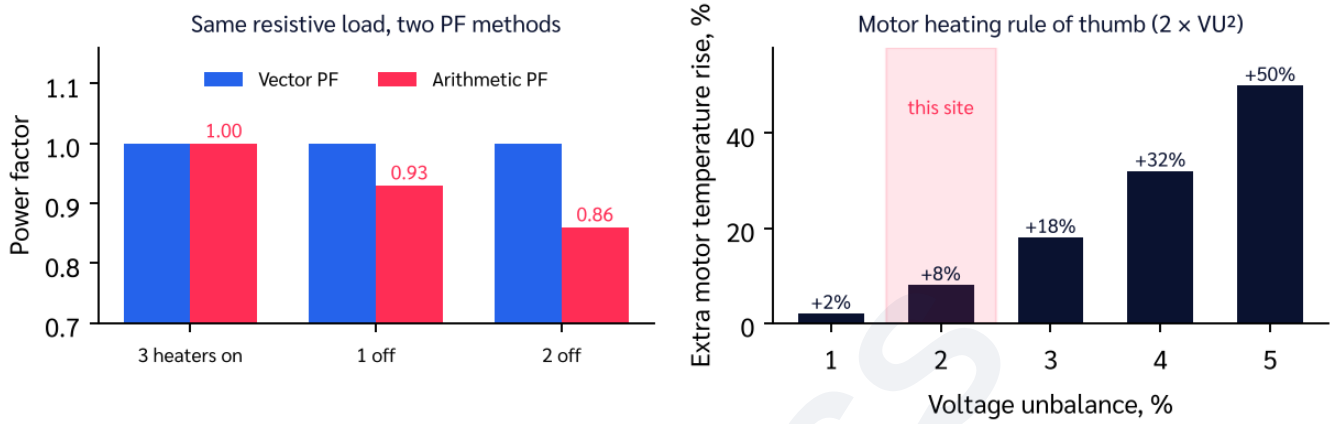
Claim	Verdict	Why
Phase unbalance lowers power factor.	Depends on the definition	With vector PF (total kW ÷ total kVA, IEEE 1459), unbalance alone does not lower PF. With arithmetic PF (sum of each phase's kVA), the same resistive load reads 0.93 or 0.86. Adding capacitors to "fix" that would increase losses.
Unbalance causes utility PF penalties.	Not by itself	PEA charges 56.07 baht per kVAR above 61.97% of kW demand, measured on the three-phase total. The charge comes from reactive loads, not from unbalance.
Single-phase-sensing PFC controllers misjudge an unbalanced site.	True	Many capacitor-bank controllers sample one phase only, so they can over- or under-compensate. It matters only where a capacitor bank exists.
Unbalance creates negative-sequence current and extra heat.	True for voltage unbalance	Motors heat by about 2 × (% voltage unbalance)², and current unbalance in a motor is 6–10 times the voltage unbalance.
Active filters or static var generators are the solution.	Only when measured	They are justified only by real kVAR charges or measured harmonics. Load redistribution comes first.

2 What the meter shows at this site



Voltage unbalance (NEMA method: largest deviation from the average phase voltage) stays near 1% at night but rises to 1.7–2.0% on average during working hours, with 5-minute peaks of 4.5–5.2%. The heavy phase, L1, sags most: on 17 October it fell to about 223 V in the afternoon while L2 and L3 stayed near 233 V. The site's meter records 0 kVAR and no power factor, so reactive power cannot be judged from the monitoring data.

3 Two effects, illustrated



Left: in a test with purely resistive heaters, switching heaters off made the load unbalanced. Vector PF stayed at 1.00, while arithmetic PF dropped to 0.93 and 0.86, with no reactive power present. **Right:** the US Department of Energy rule of thumb. At 2% voltage unbalance a motor's temperature rise increases by about 8%, roughly 8 °C on a 100 °C rise, and insulation life halves for every 10 °C. Keep voltage unbalance at motor terminals at or below 1%.

4 Standards and thresholds

Item	Limit	Source
Voltage unbalance at motor terminals	≤ 1%	NEMA MG1 / US DOE
Voltage unbalance of the supply	≤ 2% (95% of 10-minute values)	EN 50160 / IEC 61000-2-2
Voltage unbalance, utility supply	≤ 3% (no load)	ANSI C84.1
PEA power factor charge	56.07 baht/kVAR above 61.97% of kW demand (about PF 0.85)	PEA tariff
Current unbalance in a motor	6–10 × the voltage unbalance	US DOE

5 What to do

1. **Rebalance single-phase loads** No cost
   Move about 1–1.5 kW of daytime single-phase load from the heavy phase (L1) to the light one (L2). This reduces current and voltage unbalance together.
2. **Check the PEA bill for a kVAR charge** No cost
   If there is none, power factor is not a cost problem. If there is, measure which loads cause it.
3. **Check any capacitor bank** Site visit
   If the site has one, make sure its controller senses all three phases, and that its steps match the load.
4. **Log voltage unbalance at critical motors** Measurement
   A one-week power-quality log at the machine terminals. Above 1%, derate motors or fit phase-unbalance protection relays.
5. **Buy SVG or active filters only on evidence** Investment
   Only if the bill shows real kVAR charges or measurements show high harmonics.

In one sentence: phase unbalance is a load-distribution problem first. It raises voltage unbalance and heats motors, but it does not create a power factor charge by itself. Rebalance the loads, then measure before spending on power-quality equipment.

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Prepared by Solar ESS for general information, 30 October 2026. Site figures come from SolisCloud meter data and are estimates; a power-quality analyser gives definitive values. Vendor case-study figures are not independently verified.

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